Project Update

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Agenda

- July 7-14 recap
  - Design Goal
  - Concept Selection
  - CAD
- CFD Progress
- AI
- Next Steps & Questions

July 7-14 Recap

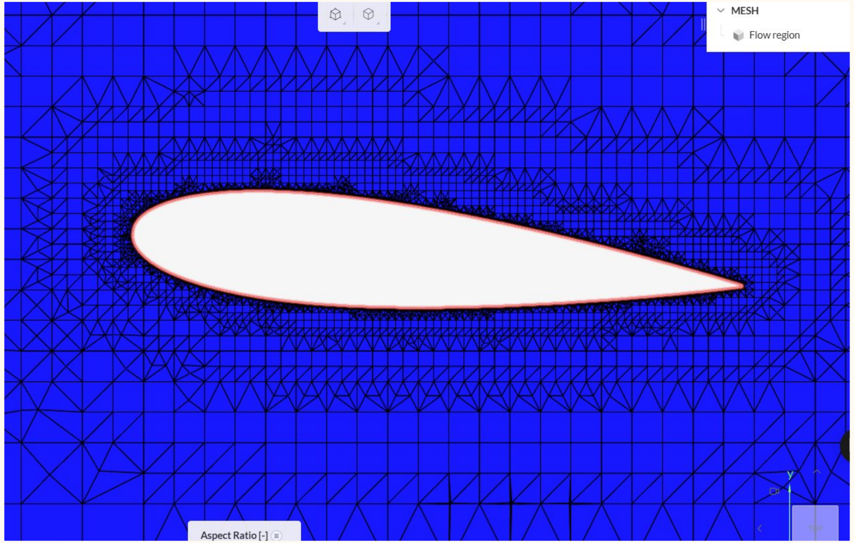
- Used LLM to generate project timeline
- Determined project design goal → BOS airport
- Established design criteria to inform concept selection
- Used LLM wiki for accelerated concept selection process
    - Human + LLM selected designs
    - Ranked and used weighted scoring to pick from top choices
    - LLM scored all designs in wiki and used different weighting schemes to ensure validity
- 90% completed baseline CAD model of selected design

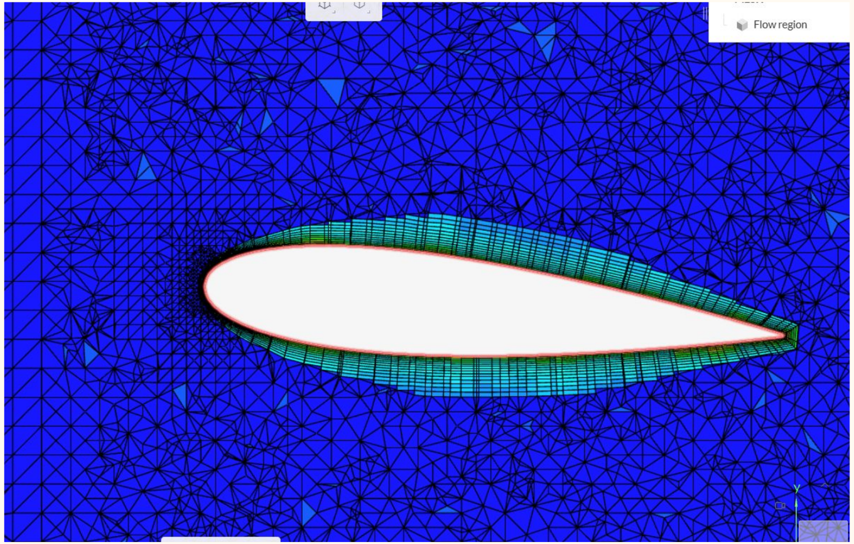
July 21 Recap (AI)

- Very helpful for initial CFD learning (walkthrough of SimScale setup, quizzes on CFD concepts, uses, limits, and reliability)
- Fusion AI for CAD produced a good VJ20 model in about four minutes, under one-third the time ZooKeeper took, with comparable or better initial quality
- AI-assisted research identified a published NACA 0012 validation approach that motivated a 2D model; this substantially improved the reported lift result
- LLM summarized daily documentation and CFD iterations and pulled out important events and themes
- LLM wiki (and other AI) utility plateaued
- LLM misreported cut in speed for top design during selection process

CFD Progress (July 14-28)

- Oriented the wiki towards CFD
    - Almost all SimScale documentation, including forum
    - Papers with well-documented airfoil and vawt cfd studies
- Used the LLM to set up simulations, and to learn about specific settings
- Working on an airfoil validation study
    - Set simulation settings for flow similarity with AirFoil tools data
    - Compared NACA airfoils 0010, 0012, 0016, 0018
    - Met w/ Phillip, learned cfd workflow best practices and mesh importance
    - Iteratively improved mesh quality


MESH
Flow region
Aspect Ratio [-]


Flow region

AI in the CFD Process

- We have been using the LLM, Gemini, and Ray (SimScale’s built in chatbot)
  - The LLM was very helpful for initial set up an learning, but quickly got repetitive
  - Gemini was most helpful for visual assessments on the mesh quality
    - Phil recommended we gave it screenshots and ask it to inspect
  - Ray is best for specific setting questions, but although it is integrated in simscale it can not read settings or verify mesh quality
    - Does link helpful pages and gives supporting images
- AI has been extremely helpful for initial set up but has somewhat plateaued in its utility as settings have become more complex

Next Steps

- Finish airfoil validation (by EOD Wednesday)
- Begin simulating base VAWT design (by Monday)
- Select design parameters of interest (by Monday)

Supplementary slides for catch up →

Design Goal

Proposed location: Governor’s Island, Boston Logan Intl. Airport

- Abundantly available wind data with very good resolution
- Avg wind speed ~5 m/s
- High energy demand
- Demonstrated interest in reducing carbon emissions

Design Goal

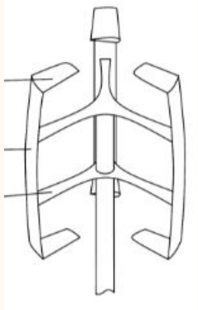
- Wind data collected at 9 m high, center of airfield (yellow star)
- Proposed turbine center rotor height 9 m (total height ~ 10 m)
- FAA regulations require distance of ~200 m from runway
- Proposed turbine location on Governors Island (red star)
  - ~ 10 acres of non-restricted space

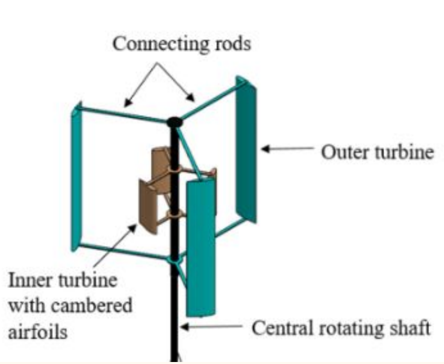
Design Criteria

- Optimized for installation site
  - Low cut in speed to take advantage of available wind power
  - Rated speed and cut out speed matching BOS wind conditions
- Economic feasibility
  - Goal: offset its cost within half its lifetime
- Comply with airport regulations
- Not more complex than necessary
- Minimal maintenance requirements
- Feasible for our project timeline

Concept Selection

Rank	Candidate	Weighted score (/5)	Notes
1	Self-starting straight-bladed H-rotor Darrieus	4.40	Highest overall mainly because it stays strong in both startup and efficiency, which dominate your weighting.
2	vj20 Proposed Hybrid VAWT	4.25	Very close second because of its top efficiency score and strong startup score, now nearly matching the straight-bladed H-rotor path.
3	vj9 Scooplet-Based Savonius	3.08	CAD score helps a lot here, but low efficiency and rated-speed fit keep it mid-pack.
4	va20 Involute Rotor with Wind Flow Modifier	3.05	Good efficiency and decent growth potential, but weak rated-speed fit and low CAD score reduce the total.
5	Low-TSR helical Darrieus	2.88	Good rated-speed fit, but the lower startup and efficiency scores hurt because those categories carry most of the weight.
6	va23 50% STS-VAWT	2.48	The weak startup score hurts most under this weighting, even though CAD and cut-out fit are better.

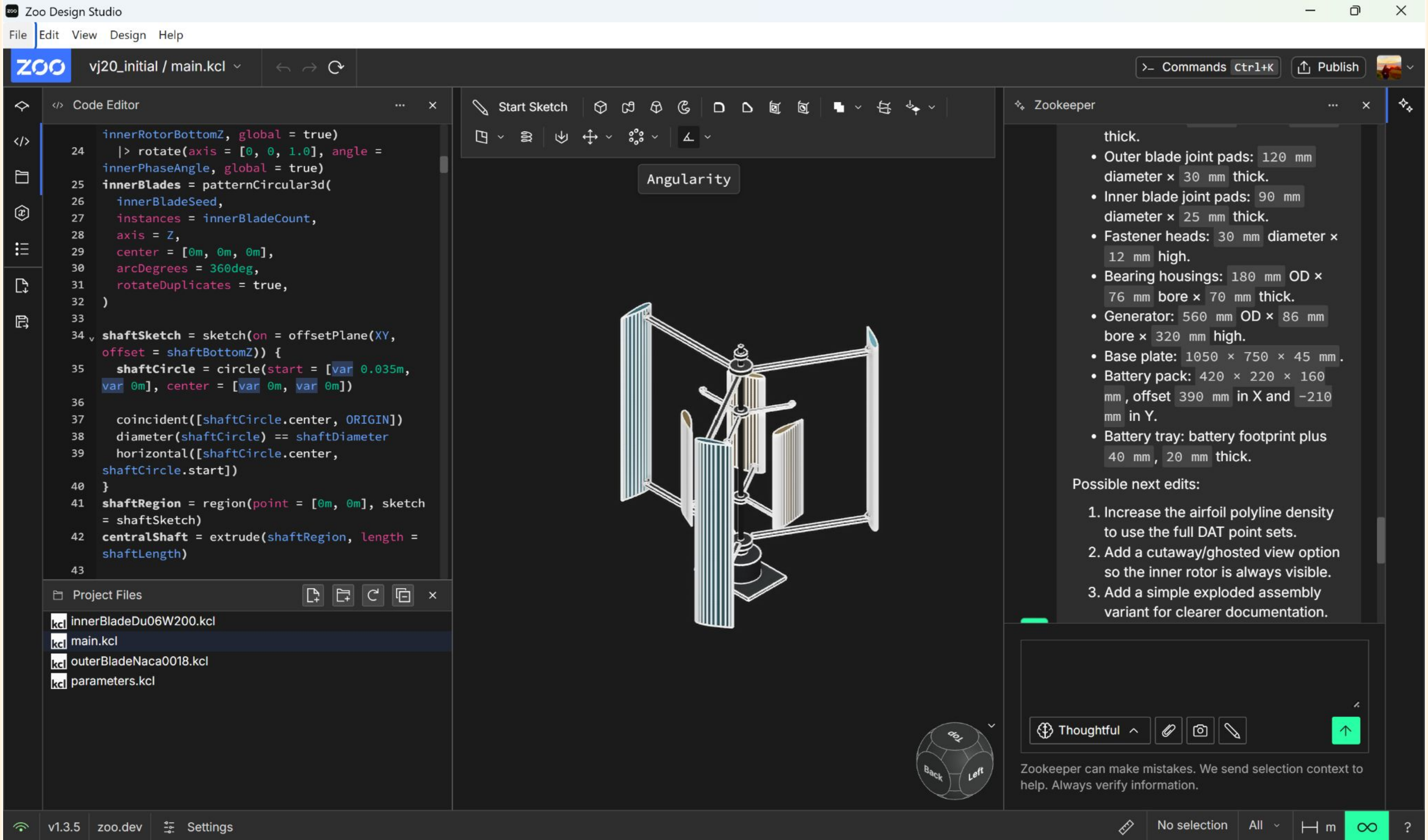




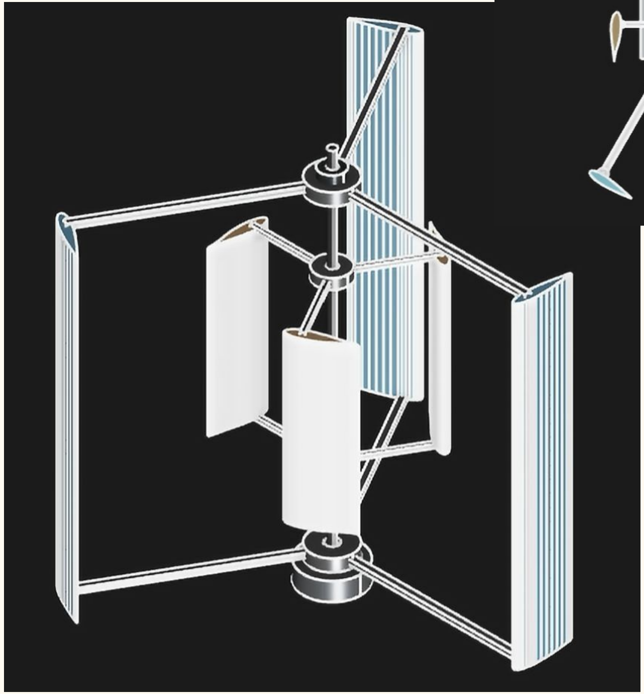
Reflections

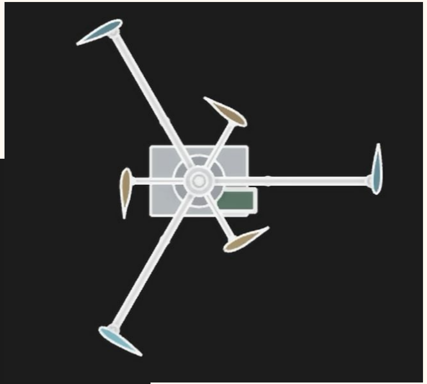
- LLM wiki *drastically* reduced the time needed for concept selection
- Very helpful to instantly pull out specific information across our ~60 sources and compare findings from many studies
    - Chat interface makes it much easier to find what you need even if you have limited knowledge about the content

- Removed the opportunity for deep research and learning about the vawt concepts and designs
    - No extended time to think about the options, decision was made in a day
- Results feel a little bit arbitrary and because of our own limited knowledge we can’t feel entirely confident about the decisions


Zoo Design Studio
File Edit View Design Help
vj20_initial / main.kcl
Commands Ctrl+K
Publish
Code Editor
innerRotorBottomZ, global = true)
|> rotate(axis = [0, 0, 1.0], angle =
innerPhaseAngle, global = true)
innerBlades = patternCircular3d(
innerBladeSeed,
instances = innerBladeCount,
axis = Z,
center = [0m, 0m, 0m],
arcDegrees = 360deg,
rotateDuplicates = true,
)
shaftSketch = sketch(on = offsetPlane(XY,
offset = shaftBottomZ)) {
shaftCircle = circle(start = [var 0.035m,
var 0m], center = [var 0m, var 0m])
coincident([shaftCircle.center, ORIGIN])
diameter(shaftCircle) == shaftDiameter
horizontal([shaftCircle.center,
shaftCircle.start])
}
shaftRegion = region(point = [0m, 0m], sketch
= shaftSketch)
centralShaft = extrude(shaftRegion, length =
shaftLength)
Project Files
innerBladeDu06W200.kcl
main.kcl
outerBladeNaca0018.kcl
parameters.kcl
Start Sketch
Angularity
Zookeeper
thick.
Outer blade joint pads: 120 mm diameter × 30 mm thick.
Inner blade joint pads: 90 mm diameter × 25 mm thick.
Fastener heads: 30 mm diameter × 12 mm high.
Bearing housings: 180 mm OD × 76 mm bore × 70 mm thick.
Generator: 560 mm OD × 86 mm bore × 320 mm high.
Base plate: 1050 × 750 × 45 mm.
Battery pack: 420 × 220 × 160 mm, offset 390 mm in X and -210 mm in Y.
Battery tray: battery footprint plus 40 mm, 20 mm thick.
Possible next edits:
1. Increase the airfoil polyline density to use the full DAT point sets.
2. Add a cutaway/ghosted view option so the inner rotor is always visible.
3. Add a simple exploded assembly variant for clearer documentation.
Thoughtful
Zookeeper can make mistakes. We send selection context to help. Always verify information.
Top
Back
Left
v1.3.5
zoo.dev
Settings
No selection
All
m

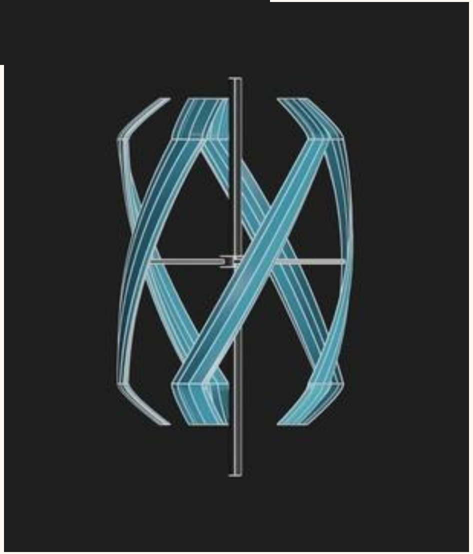
CAD





← Hybrid H-type (vj20)

EN0005 H-type (va9)→



Design Selection Conclusion

- Hybrid H type was easily modeled and has very promising performance specs
  - Always ranked in top designs during selection process
- EN0005 H type was more challenging to model
- EN0005 was frequently winning comparisons due to low cut in speed
  - Difficult to test in SimScale
  - Potentially difficult to recreate w/out design specs
  - Not necessarily the most important spec for BOS site

- Initial thought: combine design strengths by applying EN005 H type unique geometry to the Hybrid H type turbine
  - Airfoil
  - Blade end positions